SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: Artificial Intelligence **COURSE CODE: CSA17**

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. *(BL3)*

Assessment Tool Weightage: 50%

QUESTIONS: 5

Duration : 1 Hour
Total Marks:50

Annexure A

Analytical Problem Solving

Code of Conduct:

*I **Thamaraiselvi S (192511042)** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Thamaraiselvi S

Annexure A

Analytical Problem Solving

1. Solve the Water Jug problem: you are given 2 jugs, a 4-gallon one and 3-gallon one. Neither has any measuring maker on it. There is a pump that can be used to fill the jugs with water. How can you get exactly 2 gallons of water into 4-gallon jug? Explicit assumptions: A jug can be filled from the pump, water can be poured out of a jug onto the ground, water can be poured from one jug to another and that there are no other measuring devices available.

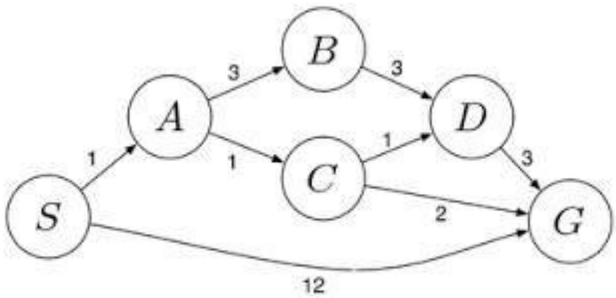
2. Find out how Mars Rover, analyse and explore the samples on Mars by answering the following questions.
   i. What are the percept's for this agent?
   ii. Characterize the operating environment.
   iii. What are the actions the agent can take?
   iv. How can one evaluate the performance of the agent?
   v. What sort of agent architecture do you think is most suitable for this agent? Why?

3. Explore the search space using search strategies and formulate the problem components for the 8-Queens problem using the following information: Place 8 queens on a chessboard such that none of the queen's attack any of the others. A configuration of 8 queens on the board as shown in the figure below, but this does not represent a solution as the queen in the first column is on the same diagonal as the queen in the last column.

4. A customer wants to travel from the one location to another location using OLA Cab booking mobile application. The customer can select the any of the cab type as mini, micro, sedan, shared, prime and so on based on his comfort to reach the destination. Identify what type of problem-solving agent can be used and write the pseudocode for the above problem.

5. A logistics company operates a delivery network where packages must be transported between warehouses at minimum cost. Each route between warehouses has an associated travel cost (fuel + time). The company wants to determine the least-cost path from the starting warehouse **S** to the destination warehouse G using the Uniform Cost Search (UCS) algorithm.

   The delivery network is represented as a weighted graph:



RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understandin g	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation
Problem Understandin g	20%	Clearly interprets problem, identifies all parameters and	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem

		requirements accurately			

1. Water Jug Problem

Problem Statement:

There are two jugs:

- Jug A = 4 gallons
- Jug B = 3 gallons

The objective is to obtain exactly **2 gallons in the 4-gallon jug**.

State Representation:
State = (A, B)
Where:

- A = Water in 4-gallon jug
- B = Water in 3-gallon jug

Initial State = **(0,0)**
Goal State = **(2,x)** where x can be 0–3.

Operators:

- Fill Jug A
- Fill Jug B
- Empty Jug A
- Empty Jug B
- Pour A → B
- Pour B → A

Solution:

Step	Action	State (4,3)
1	Fill 3-gallon jug	(0,3)
2	Pour into 4-gallon jug	(3,0)
3	Fill 3-gallon jug again	(3,3)
4	Pour into 4-gallon until full	(4,2)
5	Empty 4-gallon jug	(0,2)
6	Pour remaining 2 gallons into 4-gallon	(2,0)

Goal Reached:

The 4-gallon jug contains exactly 2 gallons.
AI Technique Used:

- State Space Search
- Problem Solving Agent
- Breadth First Search (or) Depth First Search

2. Mars Rover Agent

PEAS Description

Performance Measure:

- Collect maximum scientific samples
- Avoid obstacles
- Minimize energy consumption
- Send accurate information to Earth
- Complete mission safely

Environment:

- Mars surface
- Rocks
- Sand dunes
- Mountains
- Craters
- Dust storms

Actuators:

- Wheels
- Robotic arm
- Drill
- Camera
- Communication antenna
- Solar panels

Sensors:

- Navigation camera
- Hazard camera
- Spectrometer
- Temperature sensor
- Soil analyzer
- Battery sensor

(i) Percepts:

- Rock images
- Soil composition
- Atmospheric pressure
- Temperature
- Battery status
- Position
- Obstacles

(ii) Operating Environment:

Property	**Description**
Observable	Partially Observable
Agents	Single Agent
Deterministic	Mostly Deterministic
Dynamic	Dynamic
Discrete/Continuous	Continuous
Known	Partially Known

(iii) Actions:

- Move
- Stop
- Turn
- Capture image
- Drill rock
- Collect sample
- Analyze sample
- Upload data
- Recharge battery

(iv) Performance Evaluation:

- Number of successful missions
- Distance covered
- Accuracy of sample analysis
- Battery efficiency
- Number of obstacles avoided
- Successful communication

(v) Suitable Agent:

Learning + Utility-Based Agent
Reason:

- Learns from previous experiences.
- Makes intelligent decisions.
- Optimizes battery usage.
- Chooses safest path.
- Adapts to unknown environments.

3. 8-Queens Problem

Problem Formulation

Initial State:
Empty chessboard.

State Space:
All possible placements of queens.

Operators:
Place one queen in one column.

Goal Test:
No two queens attack each other.

Path Cost:
Each move costs 1.

Constraints:
No queens should share

- Same row
- Same column
- Same diagonal

Mathematically
For any queens i and j
$Row(i) \neq Row(j)$

$Column(i) \neq Column(j)$

$|Row(i)-Row(j)| \neq |Column(i)-Column(j)|$

Search Strategies:

- Backtracking
- Depth First Search
- Constraint Satisfaction Problem (CSP)

Advantages:

- Reduces unnecessary search.
- Efficient.
- Guarantees a valid solution.

4. OLA Cab Booking

Problem Formulation

Initial State:
Customer opens OLA application.

Goal State:
Customer reaches destination.

State Space
Available cabs:
- Mini
- Micro
- Sedan
- Prime
- Shared
- Auto

Operators:
- Select pickup
- Select destination
- Choose cab
- Book cab
- Start ride
- Complete payment

Performance Measure:
- Minimum waiting time
- Lowest fare
- Fastest route
- Safe ride

Type of Agent
Goal-Based Agent
Reason:
- Goal is reaching destination.
- Compares different options.
- Chooses best cab based on user's preference.

Pseudocode:

BEGIN

Start App

Enter Pickup

Enter Destination

Search Nearby Drivers

Display Available Cabs

User Selects Cab

Estimate Fare

IF Driver Available

Assign Driver

Display Driver Details

Start Trip

Reach Destination

Calculate Final Fare

Make Payment

Display "Ride Completed"

ELSE

Display "No Cab Available"

ENDIF

END

Advantages:

- Easy booking
- Time saving
- GPS tracking
- Online payment
- Multiple cab choices

5. Uniform Cost Search

Problem Formulation

Initial State:
Warehouse **S**

Goal State:
Warehouse **G**

State Space:
All warehouses.

Operators:
Move through connected routes.

Cost Function:
Fuel cost + Travel time.

Uniform Cost Search Steps:
1. Insert Start node into Priority Queue.
2. Remove node having lowest cost.
3. Expand neighboring nodes.
4. Update cumulative path cost.
5. Continue until Goal is reached.
6. Return minimum-cost path.

UCS Pseudocode:

UniformCostSearch(Graph, Start, Goal)

Priority Queue ← Start

Visited ← Empty

WHILE Queue is not Empty

Current ← Remove Lowest Cost Node

IF Current = Goal

Return Solution

ENDIF

FOR Each Neighbor

New Cost = Current Cost + Edge Cost

IF Neighbor not visited

Insert Neighbor

ENDIF

ENDFOR

ENDWHILE

Advantages:

- Complete algorithm.
- Always finds the least-cost path.
- Works for weighted graphs.
- Suitable for logistics and route optimization.

Disadvantages:

- Requires more memory.
- Slower than BFS when costs vary.
- Expands many nodes before reaching the goal.